SW Engineering CSC648-848 Spring 2026

Gator Freighter

Team 14

Role	Name
Team Lead	Leslie Garcia, lgarcia26@sfsu.edu
Front-End Lead	Jordan Westover
Back-End Lead	Stiofan Condon
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Milestone 1

Date Submitted	March 20, 2026
Date Revised	TBD

1. Executive Summary:

College students will be able to buy, sell, and trade everyday items from textbooks to electronics to furniture and more, with Gator Freighter. This online application will ensure that students can rely on quick, affordable, and safe transactions. General platforms like Facebook Marketplace or Craigslist expose students to strangers, unreliable transactions, and inconvenient or unsafe meetup locations. At San Francisco State University, where the cost of living and education is already a significant burden, there is now a clear, hassle-free, campus-focused solution with Gator Freighter.

Gator Freighter is a web-based marketplace exclusively for the SFSU community. The platform makes transactions simple and intuitive: users browse listings, connect with sellers through a built-in messaging system, and coordinate exchanges at designated safe meetup locations. Every user is verified through their SFSU emails, meaning every person on the platform is a member of the university community. This design is also made to reduce the time and effort students spend searching for relevant items, as users are presented with items that reflect common student needs. This creates a layer of trust and accountability that no general-purpose marketplace can offer to students.

Beyond improving individual transactions, Gator Freighter will make it easier for students to support each other, strengthening the SFSU community. Instead of relying on outside platforms, students can keep resources within their own campus network, helping others save money while building trust and familiarity. This will create a more connected environment where students aren't just users of a marketplace, but active participants in a shared system that benefits the whole campus.

Gator Freighter is being developed by a student startup team from San Francisco State University's Computer Science department. As active members of the SFSU community, we understand firsthand the frustrations of overpaying for textbooks, struggling to furnish an apartment on a tight budget, or simply trying to find a buyer for last semester's lab kit. We are building the practical solution we wish existed when we arrived on campus. We believe Gator Freighter will create lasting value to SFSU's community and for those who come after us.

2. Personae:

Persona 1: Alex

Alex is an undeclared SFSU student that has just moved into the dorms and is looking for furniture. He needs an affordable desk, lamp, and additional storage furniture, along with other dorm essentials from other students who may be in their last year. He's hoping to browse listings relevant to student living to quickly find what he needs and pick up the items locally. He isn't sure what he's studying yet but he knows he's not pursuing Computer Science since he has trouble navigating apps.

Persona 2: Tom

A graduate of SFSU named Tom, a student that is planning on moving out of the dorms and is selling furniture. He has a bedframe, desk, a standing lamp, and textbooks that have no use for him. Tom wants to make sure the app is safe from scammers and is trustworthy. Tom prefers selling to other SFSU students because it feels easier and safer than dealing with strangers on platforms like Craigslist or Facebook Marketplace.

Persona 3: Jan

Jan is a chemistry major in her third year at SFSU who is taking an intense chemistry lab that requires some materials like safety goggles and beakers. She needs something quick and nearby because her assignment is due next week! She's hoping the app isn't one of those that take 20mins to sign up for.

Persona 4: Alice

Alice was a biochemistry major, who has now switched majors to communications and is looking to get rid of lab materials she no longer has use for. She has experience using Facebook marketplace and Craigslist for her own hobbies. She is willing to sell it as cheap as possible, Alice is looking to rid herself of any reminder of the organic chemistry class she took.

3. High-level Use cases:

Use Case 1: Browse and Search Listings:
Alex is an incoming freshman who has just moved into SFSU dorms. With a limited budget and no transportation, he needs affordable furniture and other dorm stuff from other students on campus. He visits the platform and begins browsing available listings displayed on the homepage. Using the search and filtering tools he then narrows results by category until he finds items that he is looking for. After reviewing descriptions and images of the items, he then decides whether to contact the seller.

Use Case 2: Create and Manage Listings:
Tom is graduating from SFSU and is going to move out of his dorm at the end of the semester. Since he no longer needs his furniture and textbooks and other lab materials, he decides to sell them before leaving campus. He makes an account with Gator Freighter and navigates to the listing page. He then enters item details, uploads photos, sets a price, and selects whether the item is for sale or trade. After making his listing, it becomes visible to other SFSU students browsing the platform. He likes that there was verification that only other SFSU students and/or faculty can use this app while creating his account.

Use Case 3: Buyer
Jan just got into her chem class from a waitlist. She needs to buy a chemistry textbook imminently as the class is already two weeks in. She goes to Gator Freighter and searches for the book. She is prompted to create an account when she tries to message the seller. After creating an account, with her SFSU email, she begins messaging with the seller to arrange a meet. The seller asks if she would like to meet at one of the safe transaction areas on campus. She's glad she could find it in such short notice.

Use Case 4: Responding to buyers
Alice is moving out and needs to sell her biochemistry lab materials quickly. She makes an account and lists her lab materials at a competitive price. After noticing limited engagement, she logs into her account dashboard and edits the listing to lower the price and update the description. She soon begins receiving messages. She goes to her account and is able to view messages and respond.

Use Case 5: Trader
Daniel is already an avid Gator Freighter user. He is trying to avoid spending money and would rather exchange items he already owns for something he needs. He wants to trade his MacBook for a gaming laptop. He searches for a listing that matches his needs and is willing to propose a trade through the platform. To initiate the trade, he communicates via our built in messaging tool to make the proposal. After they accept, they decide a location for the campus meetup from a provided list to complete the exchange.

4. List of main data items and entities – data glossary/description

- Storefronts: The homepage of a seller's listings, showcasing their items.
- SFSU-verified access (User): A registered student with a valid sfsu email who can create listings, send messages , and participate in trades. Users have different permission levels.
- Location-based meetup: A designated campus-based meetup location selected by users to complete a trade or sale.
- Listing: A post created by a user offering an item for trade or sale. A listing includes information such as title, description, category, condition and availability status.
- Trade Request : A proposal made by a student to exchange items with another student. A trade request links two listings and requires approval from both parties before completion.
- Message: A communication between users regarding a listing or trade. Messages allow users to negotiate details and fix meetups.
- Report/complaint : A flag to a listing by a user to another user for violating platforms rules and standards.
- Course: An SFSU course identifier (CHEM 215…) that can be associated with the
- listings for course specific materials like lab stuff.
- Category : Organize listing based on specific categories like Electronics, Lab materials, Furniture, and other academic stuff.

5. List high level functional requirements

Unregistered users:

1. Unregistered users shall be able to browse available listings.
2. Unregistered users shall be able to search listings by keyword.
3. Unregistered users shall be able to filter listings by category.
4. Unregistered users shall be prompted to register or log in when attempting restricted actions.
5. Unregistered users shall be prompted to verify they are an SFSU student using student email

Registered Users:

6. Registered users shall be able to create listings
7. Registered users shall be able to reply to listings
8. Registered users shall be able to create trade requests
9. Registered users shall be able to view their dashboard/account?
10. Registered users shall be able to update their listings price.
11. Registered users shall be able to categorize listings under specific categories.
12. Registered users shall be able to create/manage storefront?
13. Registered users shall be able to view message threads
14. Registered users shall be able to search listings by keyword.
15. Registered users shall be able to filter listings by category.
16. Registered users shall be able to send messages to other users via listing.
17. Registered users shall be able to delete listings.
18. Registered users shall be able to filter listings by course number.
19. Registered users shall be able to report inappropriate listings or users.
20. Registered users shall be able to specify whether a listing is for sale or for trade.
21. Registered users shall be able to accept or decline trade proposals.
22. Registered users shall be able to select an available campus pickup location for meetups.

Admin Users:

23. Admin users shall be able to review reported listings.
24. Admin users shall be able to delete homepage listings
25. Admin users shall be able suspend user accounts.

6. List of non-functional requirements

1. Application shall be developed, tested and deployed using tools and cloud servers approved byClass CTO and as agreed in M0
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All our selected application functions shall be rendered well on mobile devices (no native app to be developed)
4. Posting sales information and messaging to sellers shall be limited only to SFSU students
5. Critical data shall be stored in the database on the team's deployment server.
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application shall follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used
13. No e-mail clients or chat services shall be allowed. Interested users can only message to sellers via in-site messaging. One round of messaging (from user to seller) is enough for this application
14. Pay functionality, if any (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Media formats shall be standard as used in the market today
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "SFSU Software Engineering Project CSC 648-848, Spring 2026. For Demonstration Only" at the top of the WWW page Nav bar. (Important to not confuse this with a real application). You must use this exact text without any editing.

7. Competitive analysis (functions/features only, not business or marketing analysis):

Feature	Craigslist	Facebook Marketplace	OfferUp	Our future product
Location - based meetup	+	+	+	++
Advanced searching	+	++	++	+
Trade/Swap (not only cash)	+	+	-	++
Browsing (storefronts)	+	+	+	+
Messaging	-	++	+	+
SFSU verified only access	-	-	-	++

Legend:

+ feature exists; ++ superior; - does not exist

Summary:

Our proposed trading platform differs from our competitors such as Craigslist, Facebook Marketplace, and OfferUp by focusing more on a secure, campus-specific environment catered towards communities in SFSU. While other platforms may provide safe general browsing, messaging, and broader location based meetups, they all lack the mechanisms uniquely tailored to a specific institution such as SFSU. Given the diverse interests of our students and faculty at SFSU, verified users are able to create/view **storefronts** that cater to specific clubs, majors, and other communities. To accomplish this, our product plans to introduce **SFSU-verified access**, ensuring only authenticated students and faculty can participate. Additionally, there will be an emphasis on **trade and swap** beyond cash only transactions, supporting the flexible exchanges that are already common on campus grounds. With the combination of a strong **location-based meetup** system that provides pinpointable spots, our product aims to achieve a more seamless, convenient, and community based way to buy and sell goods on campus, than our competitors.

8. High-level system architecture and technologies used:

- Server Host: AWS - EC2 t3.small
- Operating System: Ubuntu Server 24.04.4 LTS
- Database: MySQL v8.0.45 (MySQl Community Server)
- SW Component: NGINX v1.18.0
- SW Component: Node.js v20.20.0
- Backend Framework: Express.js v4.18.3
- Frontend Framework: Vue.js v3.5.30
- Process Manager: PM2 v5.4.3
- SSL Certificate: Let’s Encrypt (Certbot v5.3.x)
- Web Analytics: Google Analytics (TBD)
- Containerization Software: Docker v28.2.2, Docker-Compose v1.29.2
- SASS: v1.97.3
- Supported Browsers: Chrome v146.0.7680.81 + Safari v26.0

9. Use of GenAI tools like ChatGPT and copilot for Milestone 1

GenAI tool used: ChatGPT 5.2

Tasks used for:
Produced pictures of all the personas based on their descriptions.
Usefulness: HIGH

How it was used:
We basically copy and pasted the personas we made up into ChatGPT and asked it to create a picture of a person from these descriptions.

Examples/Prompts:
"Generate an image of a chemistry student clearly in her lab with a lab coat."

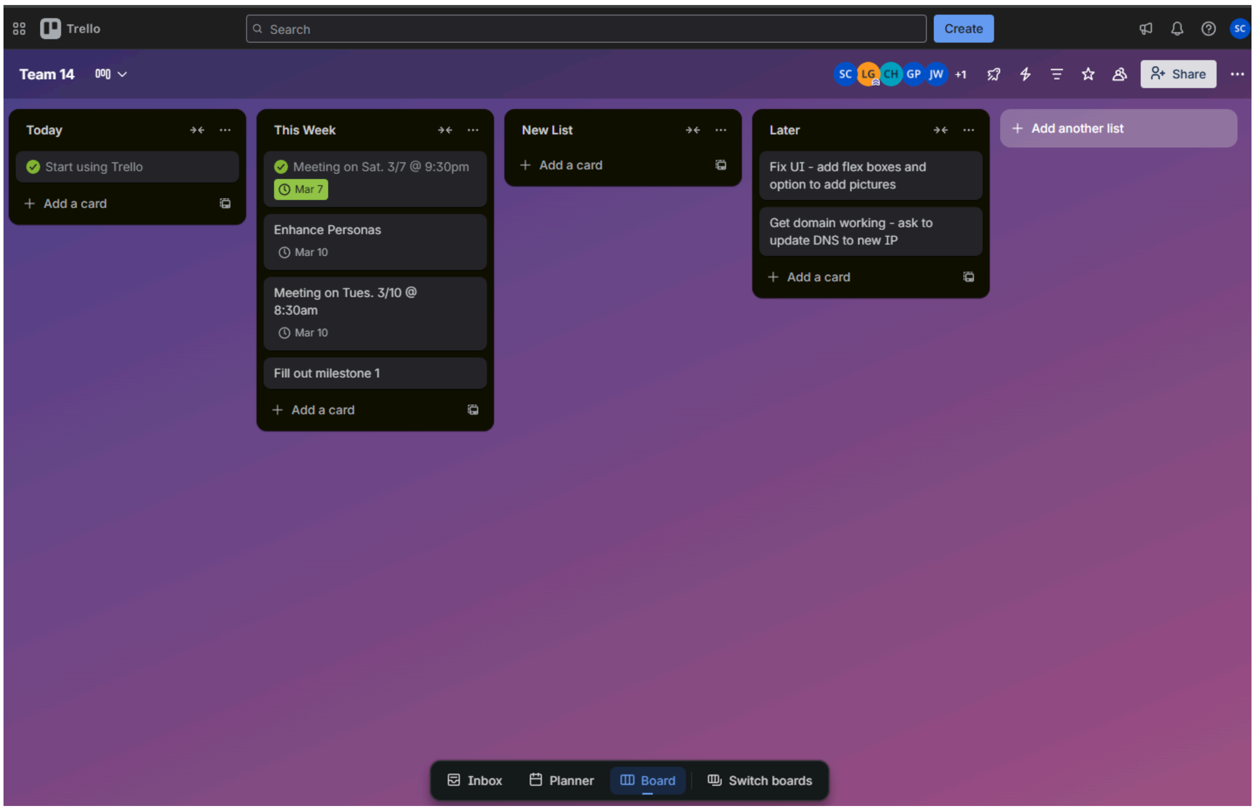
"Make an image of an SFSU student (purple shirt) browsing through a mobile listing for furniture and other items."

"Generate a few variations of an image of a mixed student choosing their major in an office."

10. Task Management Tool Selection

A: Trello

B:



11. Team and roles:

Role	Name:	Email:
Back-End Contributor	Kiran Khatri	kkhatri@sfsu.edu
GitHub Maintainer	Gursimran Grewal	ggrewal3@sfsu.edu
Front-End Lead	Jordan Westover	jwestover@sfsu.edu
Full-Stack Contributor	Christopher Huynh	chuynh3@sfsu.edu
Back-End Lead	Stiofan Condon	scondon2@mail.sfsu.edu
Team Lead	Leslie Garcia	lgarcia26@sfsu.edu
Front-End Contributor	Gabriel Purizaca	gpurizaca@sfsu.edu

12. Team Lead Checklist to be completed by team lead:

• So far all team members are fully engaged and attend team sessions when required **DONE**
• Team found a time slot to meet outside of the class **DONE**
• Team ready and able to use the chosen back and front-end frameworks and those who need to learn are working on learning and practicing **DONE**
• Team reviewed class slides on requirements and use-cases before drafting Milestone 1 **DONE**
• Team reviewed non-functional requirements from "How to start…" document and developed Milestone 1 consistently **DONE**
• Team leads checked Milestone 1 document for quality, completeness, formatting and compliance with instructions before the submission **DONE**
• Team lead ensured that all team members read the final M1 and agree/understand it before submission **DONE**
• Team shared and discussed experience with GenAI tools among themselves **DONE**
• GitHub organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.) **DONE**
• Team has chosen task management tool and invited Course instructor to project via ajsouza@sfsu.edu email. **DONE**